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Hit Finding Algorithms for the DUNE Experiment Using Single Instructions Multiple Data Parallel Processing

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The DUNE experiment will start operating at the end of this decade, with the objective of measuring in detail neutrino oscillations, and other rare physics processes. Four far detectors modules (17 kt each) will be installed at SURF, in South Dakota, about 1.5 km underground. The data selection system of one DUNE far detector module's Time Projection Chamber (TPC) relies on the real-time processing of approximately half a million channels sampled at 2 MHz. Data for each channel are analyzed to identify activity incompatible with noise (hit finding). The hits are then clustered and processed further to form a trigger decision. The aim of the DUNE data selection system is to reduce the data volume produced by the detector electronics by four orders of magnitude. In this paper, after the description of the application design, we present the implementation and tuning of multiple software-based hit finding algorithms. The high rate of incoming data (20 GB/s per host) is sustained through the employment of SiMD parallel processing, using algorithm implementations based on the AVX2 instruction set. We show the obtained performance comparing it across multiple CPU hardware platforms and conclude by illustrating live results from recent tests at a setup using prototype detector components.

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1. Introduction

The Deep Underground Neutrino Experiment (DUNE) is an international experiment aimed at probing the properties of neutrinos and exploring fundamental questions such as the matter-antimatter asymmetry in the universe. Located at the Sanford Underground Research Facility (SURF) in Lead, South Dakota, USA, DUNE employs a liquid argon time projection chamber (LArTPC) as its primary detector to observe neutrino interactions. These interactions are facilitated by a high-intensity neutrino beam generated at Fermilab, near Chicago, Illinois. LArTPC detectors are characterized by an architecture comprising a collection plane and two induction planes. This configuration facilitates precise three-dimensional reconstruction of particle trajectories. Specifically, ionization electrons drifting towards the collection plane are captured, while the induced signals are detected at the two induction planes. Figure 1 illustrates the signal formation of a typical three wire plane LArTPC. Further details on the detector technology can be found in [1].

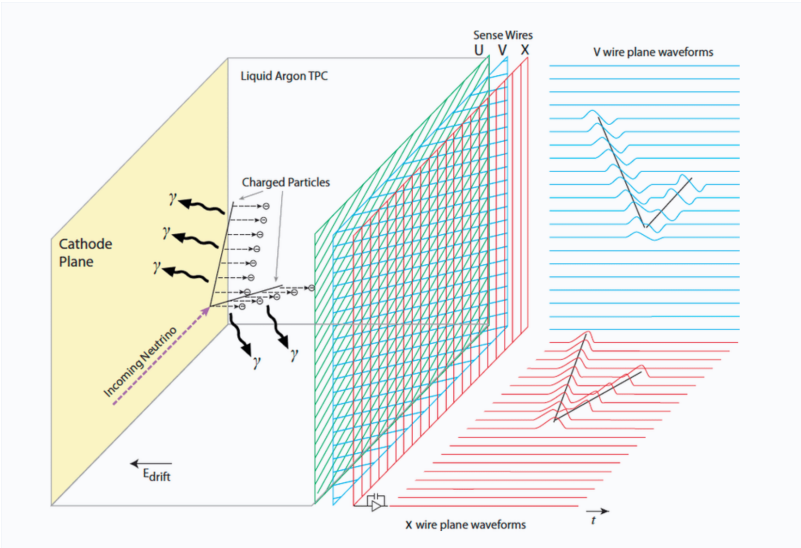


Figure 1: Schematic diagram of a three wire plane LArTPC.

The Data Acquisition (DAQ) system of the DUNE is crucial for the experiment's success because it serves as a central component, orchestrating various critical functions. Primarily, the DAQ system provides precise timing and synchronization signals to the detector electronics and calibration devices, ensuring coordinated operation and accurate data collection. One major task of the DAQ is to receive and buffer the continuous data stream from the DUNE detectors. This is followed by the application of filtering algorithms to extract relevant details, forming Trigger Decisions that identify events of interest, such as neutrino interactions. These Trigger Decisions are subsequently encapsulated into trigger records, which represent the detector data (in space and in time) associated with the identified events.

2. Hit finding

The role of the DUNE Data Selection system within the DUNE-DAQ is to identify data regions that contain events of interest within the detector raw data by using a combination of hit finding

and clustering techniques. This is done by applying different filtering steps, as illustrated in figure 2. The first step is known as Hit finding or Trigger Primitive Generation and it consists of the identification of a *Trigger Primitive* (TP) or hit of ionization from the LArTPC on a per-channel basis. Trigger Primitives are then grouped into local clusters of hits (Trigger Activity) and, if confirmed, a Trigger Candidate object is formed. The last step of the data selection process is the Trigger Decision which issues a request to permanently store the collected data.

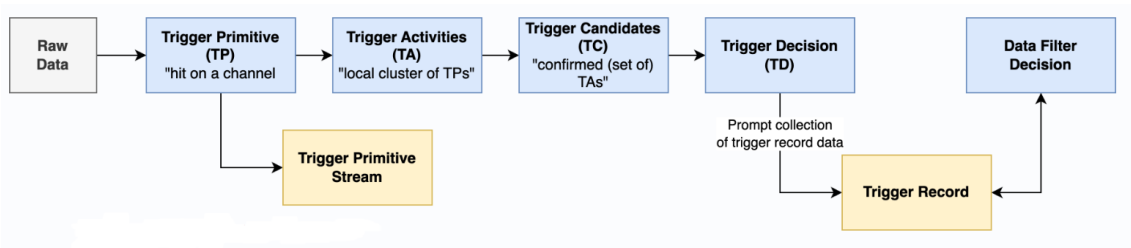


Figure 2: Diagram representing the different steps of the DUNE Data Selection system.

This paper illustrates the characterization and the performance of the first step of the data selection process: hit finding. Hit finding algorithms are crucial components for extracting information from LArTPC detectors, essential for reconstructing the signals generated by charged particles traversing the detector's liquid argon volume. These algorithms are mainly designed to identify the ionization signals induced by charged particles, known as "hits," as they drift towards the detector's readout planes. In this context, Trigger Primitives are data snippets where the signal was above a (configurable) noise-driven threshold. Trigger Primitives summarize the waveform information in terms of its threshold-crossing times, peak ADC measurements, and total ADC integral of the signal for the given start and end times.

3. Implementation of the Hit Finding

The hit finding algorithms are implemented as a set of parallel pipelines that are executed on the incoming data stream. This is motivated by two main requirements:

- support for hit finding algorithms across different detector data types
- process $\sim 200\,Gbps$ of continuous, fixed-rate TPC data from a single node

The design of the hit finding pipeline consists of multiple processing steps. Initially, the incoming data blocks are formatted by a *formatter* component. This is needed in order to adapt to the format of the incoming data stream from the electronics components. Following this step, the data are processed through multiple parallel pipelines, each executing algorithm-specific operations sequentially. The culmination of this processing chain is the identification of a Trigger Primitive object. The identified Trigger Primitives are then assembled and, subsequently, dispatched to the Trigger system for further analysis, as well as to the Dataflow system of the DUNE-DAQ for archival purposes.

Single Instruction, Multiple Data (SiMD) techniques, such as Advanced Vector Extensions (AVX), play a pivotal role in enhancing hit-finding algorithms. Leveraging AVX, which are

extensions to the x86 instruction set architecture compatible with Intel and AMD processors, enables the parallelization and augmentation of throughput during the execution of intensive algorithmic calculations. By executing the same operation (CPU instruction) simultaneously on multiple data, the use of AVX2 instructions amplifies the total processing throughput of hit finding algorithms. As a consequence, the development is then focused on optimizing memory access and overall data manipulation (bit level operations). In the current implementation of the DUNE experiment's hit-finding algorithms, 256-bit AVX2 register blocks are used for handling 16 ADC samples from the detector electronics.

Hit finding algorithms

In the context of LArTPC detectors, there are differences between collection and induction planes. Hits on the collection plane are unipolar in their form. Intuitively, hit finding on these types of signals is quite simple: apply a flat threshold that is above the noise noise and then extract the properties of the trigger primitive. On the other hand, induction wire hits are not unipolar. They instead have a positive lobe and a negative lobe. This bipolar nature of the signal is due to the charged particles passing past the wires on their way to the collection plane (see figure 1. While it is possible to run the trigger primitive on the positive lobe of the hit, a more sophisticated approach is needed for correctly identifying the full complement of the signal.

Two algorithms have been implemented in the DUNE data acquisition system: *SimpleThreshold* algorithm and the *Absolute Running Sum algorithm* (AbsRS). The SimpleThreshold algorithm is a hit finding algorithm designed to identify a signal when the incoming ADC waveform is above a configurable threshold whereas the AbsRS was designed to handle the bipolar signals from the induction planes of the LArTPC. In both algorithms, the first step consists in the formatting of the data stream from 14-bit representation to a 16-bit representation. This is followed by the live estimate of the median of the ADC waveform data in order to apply a pedestal subtraction in order to bring the waveform to zero. This step is achieved using a frugal streaming method where the estimate of the median is increased (or decreased) by one unit if the current value of the median is greater (or smaller) than the given ADC value for a fixed number of entries. Further details on the frugal streaming method can be found in [2]. In the case of the SimpleThreshold algorithm a TP object is identified if the incoming waveform (pedestal subtracted) is greater than a fixed configurable threshold. In the case of the Absolute Running Sum algorithm a further step is performed on the raw waveform, a running sum calculation is computed to specifically convert the bipolar signal from the induction planes into a unipolar signal [1]. Equation 1 shows how the absolute running sum is calculated:

$$y_n = R\, y_{n-1} + \frac{|x_n|}{s} \tag{1}$$

where x_n represents the current ADC input, y_n represents the value of the absolute running sum at sample n, s is a scale factor to avoid overflow conditions in the sum and R is factor that controls the memory of the running sum iteration.

The AbsRS algorithm essentially runs a quick integration over the raw waveform on both the collection and induction detector planes, converting the bipolar signals into unipolar signals through

[1] A similar approach is also adopted in other LArTPC experiments[3].

the use of the the absolute value of the ADC samples. Note that a second pedestal subtraction is computed on the AbsRS computation to be able to compare the the values of the AbsRS with the original waveform. As an example, figure 3 illustrates the comparison between the sample data and the result of the AbsRS computation, highlighting how the negative components of the original waveform (blue) are taken into account by the AbsRS.

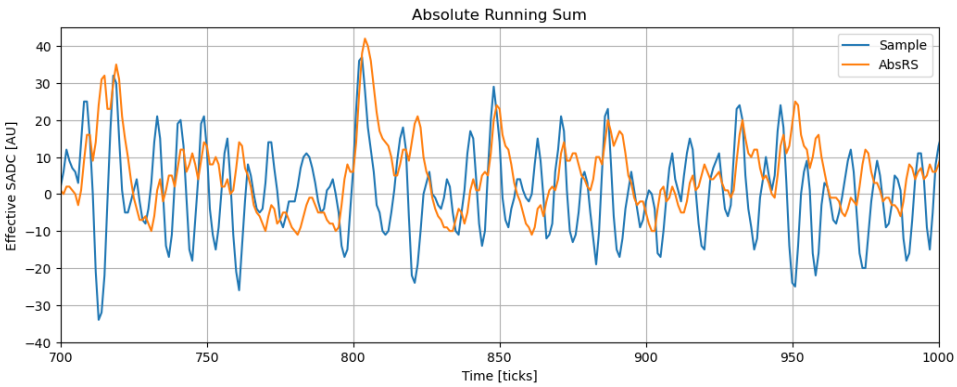


Figure 3: Comparison between original waveform sample and the AbsRS output.

4. Performance testing

The setup used for the performance testing of the Hit finding algorithms consists of *Data Producers* that emulate detector data, *Data Consumers* parse and store this information in memory and TPG threads that execute the algorithm responsible for hit finding. The testing objectives include characterizing the TPG algorithms through synthetic benchmarks to emulate Trigger Primitives (TPs), aiming to achieve TP production at the expected DUNE rate of 100 Hz per channel [1]. The testing was executed using a test application [2] running the TPG workload on a sample dataset for approximately 10 minutes (single core application). Given the algorithm's CPU, cache, and memory bandwidth-intensive nature, optimization and tuning efforts are essential to fully utilize the host machine's resources. In the testing, different strategies have been adopted to maximize the throughput, such as dedicating cores to TPG threads and maximizing hardware and data locality through software configuration to minimize context switching.

The objective of the testing is to evaluate the instructions per cycle (IPC) when running the hit finder algorithms on the same dataset sample. The IPC is a measure of a processor's efficiency in executing instructions and it indicates how many instructions a CPU can execute in a single clock cycle. This is particulary relevant when assessing the performance of algorithms that heavily rely on SiMD instructions. A higher IPC means that the CPU can execute more instructions in a given period, which generally results in better overall performance. Table 1 shows a summary of the main characteristics of the three generations of CPU used for the performance testing of the hit finder algorithms. It's important to note that, although the interest is to evaluate the algorithms in terms of IPC, other factors such as clock speed, cache size, and architecture also play a significant role in the performance evaluation.

[2] Available in tpgtools at https://github.com/DUNE-DAQ/tpgtools

Server name	**np04-srv-025**	**np04-srv-028**	**np02-srv-002**
Model name	Intel(R) Xeon(R) Gold 6136 CPU @ 3.00GHz	Intel(R) Xeon(R) Gold 6242 CPU @ 2.80GHz	Intel(R) Xeon(R) Gold 6346 CPU @ 3.10GHz
Launch date	Q3'17	Q2'19	Q2'21
Threads per core	2	2	2
Cores per socket	12	16	16
Socket	2	2	2
CPU max MHz	3700	2800	3600
L1d	768k	1 MiB	1 MiB
L1i	768k	1 MiB	1 MiB
L2	24 MiB	32 MiB	40 MiB
L3	49.5 MiB	44 MiB	72 MiB

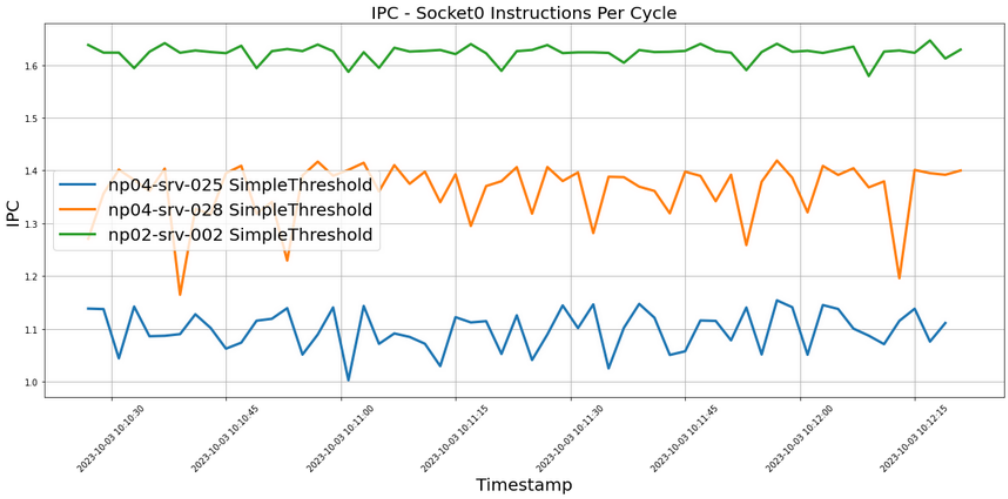
Table 1: Comparison of different CPU generations used for the TPG evaluation.

Figure 4 shows a comparison of the IPC across the three CPU hardware platforms when executing hit finder algorithms. As expected, the most recent generation of CPUs (e.g. np02-srv-002) outperform the older generations with an IPC higher than the previous generations by 15-30%.

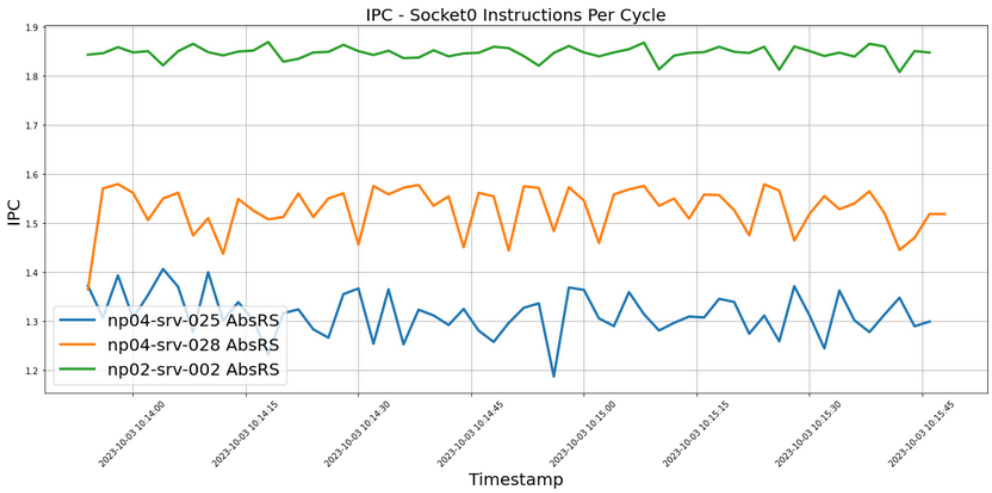
5. Results from a prototype setup

A prototype setup, also known as *coldbox*, available at CERN was used a testing platform for the hit finding algorithms. The setup is a small-scale version of LArTPC detector (figure 5) with a drift volume of approximately $\sim 30\,cm$. The advantage of using the coldbox detector is to be able to exercise hit finding algorithm on physical signals (e.g. cosmic muons, shower-like events). This allows to assess not only the performance characteristics of the algorithms but also the quality of the output Trigger Primitives. This, in turn, allows to transition from a synthetic benchmark evaluation as described in the previous section to testing on system with more realistic noise and signal conditions.

The results of the testing on the prototype setup showed that the current implementation of the hit finding algorithms can sustain the detector data rates within the resources of a single host. The testing also proved to be a useful tool to understand the performance difference between hit finding algorithms and bottlenecks oh the host machine using more realistic noise and signal conditions. As an example, figure 6 shows a track-like event (muon candidate) using the prototype setup, showcasing the channel-time view across the three detector planes. The event display illustrates that a Trigger Primitive has been identified for the hits that constitute the track. The figure also shows the ADC waveform with overlaid the identified Trigger Primitives. As it can be noted the Trigger Primitive correctly identified the peak signal above the noise level for a given hit in the candidate event.



(a) IPC when using SimpleThreshold algorithm



(b) IPC when using AbsRS algorithm

Figure 4: Result of the synthetic testing of hit finder algorithms in terms of IPC across different CPU hardware platforms.

6. Conclusion

In conclusion, this paper has presented hit finding algorithms for the DUNE experiment, which play a critical role in the Data Selection system. The study focused on implementing and optimizing these algorithms for efficient processing of the massive data stream generated by the DUNE detectors. Leveraging SiMD parallel processing techniques, particularly AVX2 instructions, the algorithms were tailored to handle the DUNE's high data rate.

Through synthetic benchmarks and testing on a prototype setup, the performance of the hit finding algorithms was evaluated, demonstrating their ability to sustain detector data rates (approximately 10 GB/s per CPU) within the resources of a single host. The testing also provided insights into the performance characteristics and bottlenecks of the algorithms under more realistic noise

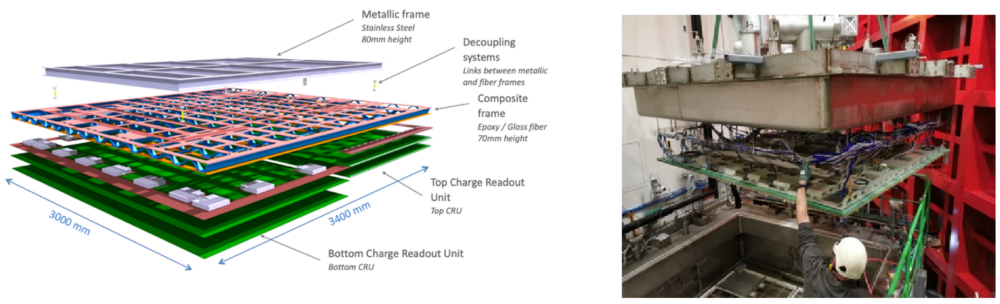


Figure 5: Schematic representing of the prototype setup used for the hit finding evaluation (left) and integration of the detector at CERN (right).

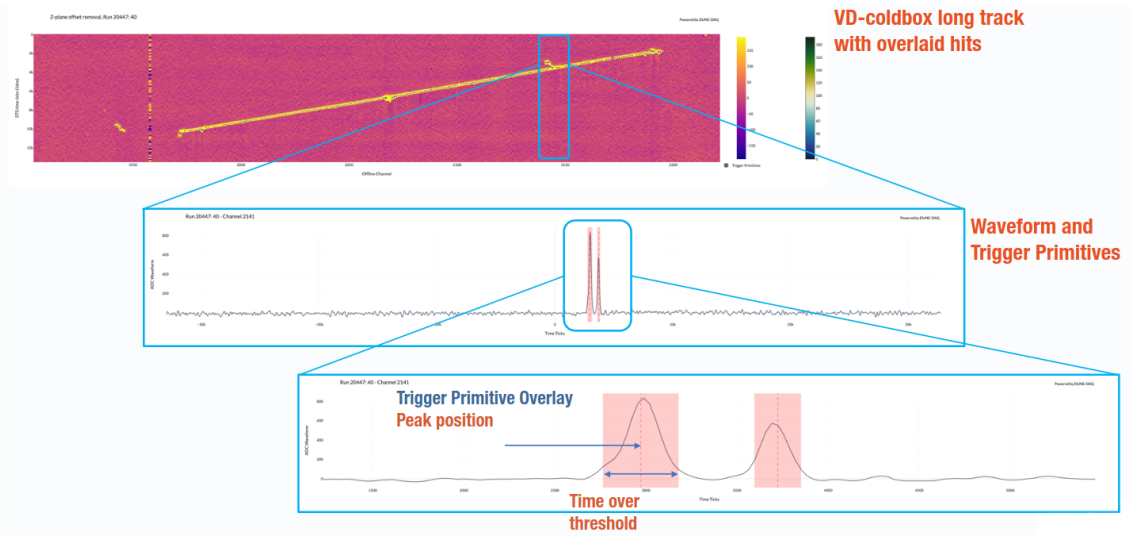


Figure 6: Example of muon candidate event and the corresponding waveform and Trigger Primitive object for the collection plane view.

and signal conditions.

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